

- 13** A stone of mass m attached to a string is whirled in a vertical circle of radius r . At the top of the circle, the tension in the string is four times the stone's weight. At this point the stone's speed is

A $\sqrt{rg}$

B $\sqrt{3rg}$

C $\sqrt{4rg}$

D $\sqrt{5rg}$

Ans: D

At the top, net force = $mg + T$

$$\frac{mv^2}{r} = mg + 4mg$$

$$v = \sqrt{5rg}$$